Multi-Channel based Light Inception with boosTing tEchniques for Time Series Classification

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***Abstract*—In recent years, time series classification (TSC) using deep learning has been actively studied. TSC is a fundamental technology for anomaly detection, prediction, and identification, and its application covers a wide range of fields including healthcare, human activity recognition, cybersecurity, and finance. Although there are a number of deep learning–based models that can accurately classify time series data, reducing the size of the model while maintaining accuracy is important for its implementation on edge devices, such as Internet of Things (IoT) devices, for the real-time processing of massive amounts of data, and for reducing power consumption. This paper proposes two lightweight models: the Multi-Channel MACD-Histogram-based LITE (MC-MHLITE) model and its ensemble variant, the Parallel Multi-Channel MACD-Histogram-based LITE (PMC-MHLITE) model. In the MC-MHLITE model, MACD histogram is integrated into the Light Inception with boosTing tEchniques (LITE) model to enhance feature extraction. Furthermore, the PMC-MHLITE model is designed as a parallel architecture comprising multiple MC-MHLITE units to further improve the versatility of the classification performance. These proposed models aim to enhance classification accuracy by leveraging many types of features while preserving the lightweight characteristic of the LITE model.**

***Index Terms*—Time series classification, Deep learning, the Light Inception with boosTing tEchniques model, style, MACD histogram**

I. Introduction

In recent years, time series data analysis has received much attention across a wide range of fields, including healthcare, human activity recognition (HAR), cybersecurity, finance, and marketing [**?**]. In time series data analysis, the task of assigning class labels to unlabeled time series data is referred to as time series classification (TSC) [**?**]. TSC is a fundamental technology for anomaly detection, prediction, and classification. Therefore, the research on developing its core, high-precision classification models is a significant challenge. In particular, time series data have their own complexities such as variations in sampling intervals, noise interference, and phase shifts over time. These characteristics make it difficult to directly apply general-purpose data classification methods to TSC.

Therefore, many classification models based on deep learning have been proposed for TSC [**?**]. In particular, convolutional neural networks (CNNs) [**?**] are an effective technique for TSC; one such representative example is the Fully Convolutional Network (FCN) model [**?**]. The FCN model efficiently extracts both local and global features from time series data using stacked one-dimensional (1D) CNNs. Additionally, the LSTM-FCN [**?**] model, which combines Long Short-Term Memory (LSTM) [**?**] with the FCN model, has also been proposed. These approaches achieve higher classification accuracy than conventional machine learning models.

Subsequently, as successors to these models, the Multi-Channel MACD-Histogram LSTM-FCN (MC-MHLF) model utilizing the MACD histogram [**?**], and its modified variants, the MC2-MHLF model and PMC2-MHLF model [**?**], have been proposed. These models calculate the MACD histogram of time series data, which represents the velocity and acceleration of the data, and treat it as multi-channel data. This emphasizes inflection points in time series data increases and decreases, enabling improved classification accuracy.

While the accuracy of these models has improved, the number of parameters in these models has increased. However, lightweight models are required for real-world applications due to real-time processing and the resource constraints of sensor and edge devices. To meet this requirement, research has focused on lightweight models, including model compression via knowledge distillation or lightweight architectures, such as Light Inception with boosTing tEchniques (LITE) [**?**], [**?**], which utilizes depthwise separable convolutions.

This study proposes a novel method that integrates the LITE model with MC-MHLF concepts to achieve both lightweight and the superior feature extraction capabilities of the MC-MHLF model. Our main contributions are as follows:

- We propose the Multi-Channel MACD-Histogram-based LITE (MC-MHLITE) model and its ensemble variant, the Parallel-Multi-Channel MACD-Histogram-based LITE (PMC-MHLITE) model. The MC-MHLITE model integrates both long-term and short-term MACD histograms into the input layer of the LITE model, using this multi-channel data alongside the input time series data. Furthermore, the PMC-MHLITE model is a parallel architecture consisting of multiple MC-MHLITE units. Each unit receives input from multi-channel data created using long- and short-term parameters, respectively.

- The UCR time series archive [**?**] was used as the evaluation dataset. In addition to classification accuracy, a comprehensive performance evaluation was conducted using the number of model parameters and processing time as metrics.

The structure of this paper is as follows: Section II discusses related work on lightweighting. Section III provides background knowledge of TSC, MACD histogram, and the MC-MHLF model. Section IV proposes new models. Section V reports the results of experiments, and Section VI summarizes the paper.

II. Related Work

With the acceleration of deep learning applications in the real world, it has become a critical challenge to implement models on edge devices and run them directly on limited computational resources. Compared to backend environments such as servers and workstations used for model training, edge devices have strict power consumption constraints and limitations on available computational power and memory capacity. Therefore, it is essential to build lightweight models that can operate under limited resources. Therefore, a significant research area has emerged: reducing the size of models while maintaining the performance of large-scale, high-precision models built on the backend.

In model lightweighting studies, there are four main approaches including pruning [**?**], quantization [**?**], knowledge distillation [**?**], and network architecture optimization.

Pruning is a technique used to eliminate redundant parameters within a neural network. Proposed methods commonly include removing connections with small absolute weights or pruning at the filter level. Quantization is a technique that reduces computational cost by converting model parameters from standard 64-bit or 32-bit floating-point numbers to lower-precision integers. Knowledge distillation is a technique that trains a small-scale model (the student model) to mimic the output of a pre-trained, high-accuracy, large-scale model (the teacher model).

Furthermore, approaches that fundamentally redesign the model architecture itself have gained significant attention. For instance, MobileNets [**?**] achieved a substantial reduction in computational complexity by introducing depthwise separable convolution [**?**]. ShuffleNet [**?**] combines channel shuffling with group convolutions to optimize the architecture for edge devices.

Although model architecture optimization was initially developed for image data, research on lightweight models is also advancing in the field of TSC. For example, in the paper [**?**], a novel model compression learning framework utilizing knowledge distillation was proposed for time series data classification models. Furthermore, the LITE model [**?**], [**?**] employs an Inception-like architecture, which incorporates parallel convolutional layers with various kernel sizes. This model achieves a significant reduction in both computational complexity and the number of parameters by employing depthwise separable convolutions instead of standard convolutions.

III. Preliminalies

A. TSC

A time series is defined as a sequence of elements recorded in time order. Let x_t denote the element at time t; a time series $\mathbf{x}$ of length n is represented as follows:

$$\mathbf{x} = \langle x_1, x_2, \ldots, x_n \rangle, \quad x_t \in \mathbb{R}$$

For multivariate time series, x_t is defined as an M-dimensional vector $x_t \in \mathbb{R}^M$, and the entire time series is described as a matrix $\mathbf{x} \in \mathbb{R}^{M \times n}$.

Time series classification (TSC) is defined as the task of predicting the class label for a given time series. Let $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N$ denote the training dataset, where $\mathbf{x}_i$ represents the i-th time series, defined as $\mathbf{x}_i = \langle x_{i,1}, x_{i,2}, \ldots, x_{i,n} \rangle$. The corresponding class label is given by $y_i \in \mathcal{Y} = \{1, 2, \ldots, C\}$, where n is the length of the time series and C is the total number of classes. The goal of TSC is to construct a classification model $f : \mathcal{X} \to \mathcal{Y}$ that predicts the class label $y = f(\mathbf{x})$ for an unlabeled time series $\mathbf{x}$.

B. MACD Histogram

The MACD (Moving Average Convergence/Divergence) histogram [**?**] is an indicator originally developed for technical analysis in financial markets, calculated using exponential moving averages (EMA). For a given time series $\mathbf{x}$, the i-th element of EMA with a window size $\omega \in \mathbb{N}$ is defined as follows:

$$ema(\omega)[\mathbf{x}]_i = \alpha \cdot x_i + (1 - \alpha) \cdot ema(\omega)[\mathbf{x}]_{i-1} \tag{1}$$

Note that x_i represents the i-th value of $\mathbf{x}$, and $\alpha = 2/(\omega + 1)$ is the smoothing coefficient. During the initial period where $i \leq \omega$, the EMA is calculated as the simple arithmetic mean as follows:

$$ema(\omega)[\mathbf{x}]_i = \frac{1}{i} \sum_{k=1}^{i} x_k \tag{2}$$

The MACD line, which represents the momentum of time series data, is calculated as the difference between two EMAs with distinct window sizes, ω_1 and ω_2 (where $\omega_1 < \omega_2$):

$$macd(\omega_1, \omega_2)[\mathbf{x}]_i = ema(\omega_1)[\mathbf{x}]_i - ema(\omega_2)[\mathbf{x}]_i \tag{3}$$

Furthermore, the Signal line is generated by applying an EMA with a window size ω_3 to the MACD values:

$$signal(\omega_1, \omega_2, \omega_3)[\mathbf{x}]_i = ema(\omega_3)[macd(\omega_1, \omega_2)[\mathbf{x}]]_i \tag{4}$$

The MACD histogram is defined as the difference between the MACD and Signal lines, serving as an indicator that captures the acceleration of changes in the data.

$$\begin{aligned} hist(\omega_1, \omega_2, \omega_3)[\mathbf{x}]_i = {} & macd(\omega_1, \omega_2)[\mathbf{x}]_i \\ & - signal(\omega_1, \omega_2, \omega_3)[\mathbf{x}]_i \end{aligned} \tag{5}$$

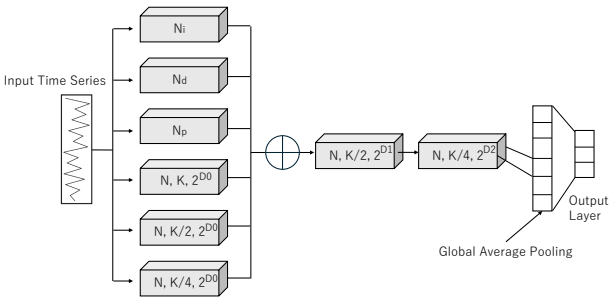


Fig. 1: LITE Model

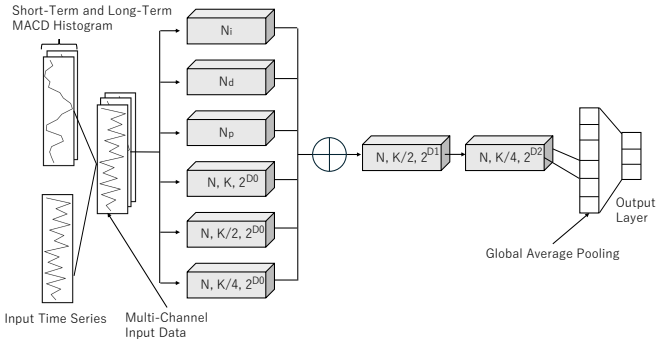


Fig. 2: MC-MHLITE Model

C. MC-MHLF Model

The MC-MHLF model combines long-term and short-term MACD histograms calculated from time series data and uses them as input as three-dimensional multichannel data. The model has a parallel architecture that consists of an LSTM unit and an FCN unit. The features extracted from both units are fused in a concatenation layer before being propagated to the output layer. The FCN consists of three convolutional layers, followed by a global average pooling (GAP) layer. To stabilize the training process and introduce nonlinearity, batch normalization and the rectified linear unit (ReLU) activation function are applied to each convolutional layer. Finally, the GAP layer extracts representative values from the feature maps as feature values and aggregates the extracted feature values into a single feature vector.

IV. Proposed Model

A. LITE Model

The LITE model is a lightweight version of InceptionTime that significantly reduces the number of model parameters. To achieve this reduction while maintaining performance, it employs boosting techniques such as standard convolutions and custom filters, in addition to depth-wise separable convolutions.

B. MC-MHLITE Model

This model (Fig. **??**) is a modification of the LITE model that accepts three inputs: the original time series data, and the short- and long-term MACD histograms extracted from the same time series data used in the MC-MHLF model. By incorporating the MACD histogram—which represents the acceleration of the time series data—into a multi-channel framework, weights are assigned to the features. Furthermore, the model emphasizes not only changes in the values of the time series data itself but also the simultaneous incorporation of the " acceleration " and trend changes of the time series data as represented by the MACD histograms. While the LITE model achieves lightweight performance by using depth-wise separable convolutions, accepting 3-dimensional multi-channel data as input enables it to learn both long-term fluctuation trends—which cannot be fully captured by a single time series —and short-term changes simultaneously. This emphasizes characteristic sub-time series and their changes, leading to an expected improvement in classification accuracy.

Here, for the time series data $\mathbf{x}$, let the MACD histogram calculated using long-term parameters be denoted as $hist(\omega_{l_1}, \omega_{l_2}, \omega_{l_3})[\mathbf{x}]$, and the MACD histogram calculated using short-term parameters as $hist(\omega_{s_1}, \omega_{s_2}, \omega_{s_3})[\mathbf{x}]$. Here, the ith element of the 3-dimensional multi-channel data is

$$multi3(\mathbf{x})_j = \langle x_i, hist(\omega_{l_1}, \omega_{l_2}, \omega_{l_3})[\mathbf{x}]_i, \\ hist(\omega_{s_1}, \omega_{s_2}, \omega_{s_3})[\mathbf{x}]_i \rangle. \tag{6}$$

C. PMC-MHLITE Model

This model (Fig. **??**) consists of two LITE models running in parallel: one that takes time series data and a long-term MACD histogram as inputs, and another that takes time series data and a short-term MACD histogram as inputs. The outputs from the Global Average Pooling layers of these LITE models are combined in a fusion layer and fed into the output layer. By combining the long-term and short-term MACD histograms with the time series data to create a multi-channel input, the aim is to process different sub-time series data using separate LITE models, thereby extracting features from each dataset without compromising their distinct characteristics. Furthermore, since the LITE model itself is a lightweight model with a reduced number of parameters, even when two models are deployed in parallel, it is believed that high-precision classification utilizing more features is possible while keeping the increase in computational cost to a minimum.

Here, for the time series data $\mathbf{x}$, let the MACD histogram calculated using long-term parameters be denoted as $hist(\omega_{l_1}, \omega_{l_2}, \omega_{l_3})[\mathbf{x}]$, and the MACD histogram calculated using short-term parameters as $hist(\omega_{s_1}, \omega_{s_2}, \omega_{s_3})[\mathbf{x}]$. Here, the ith element of the two multi-channel data sets input to PMC-MHLITE is

$$lmulti2(\mathbf{x})_j = \langle x_i, hist(\omega_{l_1}, \omega_{l_2}, \omega_{l_3})[\mathbf{x}]_i \rangle \tag{7}$$

$$smulti2(\mathbf{x})_j = \langle x_i, hist(\omega_{s_1}, \omega_{s_2}, \omega_{s_3})[\mathbf{x}]_i \rangle \tag{8}$$

is defined.

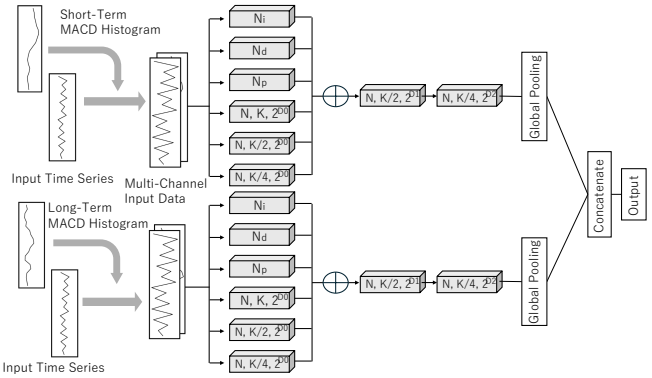


Fig. 3: PMC-MHLITE Model

V. Experiments

A. Dataset

In this experiment, we use the UCR Time Series Classification Archive released by the University of California, Riverside, which is widely used as a benchmark for time series classification. This archive consists of 85 datasets; each dataset contains time series data with varying lengths and has been pre-split into training and test datasets.

These datasets are classified into the following seven categories based on their characteristics.

- **Device**: Time series representing the states or usage patterns of electronic appliances and devices.
- **ECG**: Biometric signals captured via electrocardiography (ECG) to monitor cardiac activity.
- **Image**: Time series converted from image outlines or shapes.
- **Motion**: Data recording the physical movements and trajectories of human subjects or objects.
- **Sensor**: Physical measurements observed by environmental or industrial sensors.
- **Simulated**: Synthetic datasets generated through mathematical models or computational simulations.
- **Spectro**: Spectral data of substances (e.g., food or chemical compounds) acquired using a spectrometer.

B. Criteria

For each dataset, we trained the model using the training dataset and measured the accuracy using the test dataset. In the experiment, we compared classification accuracy, the number of parameters, and processing time with other models. The comparison models included the conventional MC-MHLF model and the LITE model.

We evaluated the models using three metrics across all 85 datasets: average accuracy, Average of Rankinging, and number of best accuracy. The average accuracy is the mean accuracy achieved for the test data across all 85 datasets. The Average of Rankinging is calculated by determining the ranking of the accuracy rates among the models for each dataset, and then averaging these rankings across the 85 datasets. The number of best accuracy is the number of datasets in which a model achieved the highest accuracy rate, after determining the ranking of accuracy rates among models for each dataset.

TABLE I: Summary of Comparison Results by Category

Category	Metric	LITE	MC-MHLITE	PMC-MHLITE
Device (6)	Average of Accuracy	**0.735**	0.728	0.725
	Average of Ranking	**1.500**	2.167	2.333
	Number of Best Accuracy	**4**	1	1
ECG (6)	Average of Accuracy	0.960	**0.963**	0.961
	Average of Ranking	**1.667**	1.833	**1.667**
	Number of Best Accuracy	**3**	2	**3**
Image (29)	Average of Accuracy	0.859	0.866	**0.869**
	Average of Ranking	2.172	**1.862**	**1.862**
	Number of Best Accuracy	10	10	**11**
Motion (14)	Average of Accuracy	0.783	0.788	**0.795**
	Average of Ranking	2.214	1.786	**1.643**
	Number of Best Accuracy	4	5	**7**
Sensor (18)	Average of Accuracy	0.862	0.861	**0.869**
	Average of Ranking	1.778	2.056	**1.333**
	Number of Best Accuracy	9	6	**13**
Simulated (5)	Average of Accuracy	0.965	0.971	**0.989**
	Average of Ranking	**1.400**	2.000	**1.400**
	Number of Best Accuracy	**4**	2	3
Spectro (7)	Average of Accuracy	**0.900**	0.886	0.897
	Average of Ranking	1.571	2.286	**1.429**
	Number of Best Accuracy	**5**	1	**5**
Total Average	Average of Accuracy	0.855	0.857	**0.863**
	Average of Ranking	1.941	1.988	**1.671**
	Number of Best Accuracy	38	26	**43**

C. Environment

Processing time was measured using a computer equipped with an NVIDIA GeForce RTX 4070 Ti (CPU: Intel(R) Core(TM) i7-4790 @ 3.60GHz, Memory: DDR3-1600 (PC3-12800) 16GB, Storage: 1TB WD Blue 3D NAND SATA WDS100T2B0A).

D. Experiment 1

Table **??** presents the results of a performance comparison between the LITE model, the MC-MHLITE model, and the PMC-MHLITE model across seven categories.

The PMC-MHLITE model achieved the highest Average of Ranking and maximum number of top-ranked results in six out of the seven categories, excluding the “ Device ” category. In particular, it achieved stable accuracy comparable to the LITE model in categories with relatively large datasets, such as “ Image, ” “ Motion, ” and “ Sensor. ” This is believed to be the result of more accurately capturing the dynamic changes in time series data by separately converting the long-term and short-term MACD histograms into multiple channels and training them in parallel.

On the other hand, in the “ Device ” category, the LITE model achieved an average accuracy of 0.735 and an Average of Ranking of 1.667, outperforming the proposed model. The datasets included in “ Device ” (such as power consumption) tend to emphasize periodicity and the presence of specific patterns, suggesting that the addition of acceleration information via the MACD histogram does not necessarily work effectively in all domains.

However, overall, the PMC-MHLITE model achieved an average accuracy of 0.863 and an Average of Ranking of 1.629, demonstrating an improvement over the LITE model

TABLE II: Summary of Comparison Results by Category (Including MC-MHLF)

Category	Metric	MC-MHLF	LITE	MC-MHLITE	PMC-MHLITE
Device (6)	Average of Accuracy	**0.755**	0.735	0.728	0.725
	Average of Ranking	**1.167**	2.333	3.167	3.333
	Number of Best Accuracy	**5**	1	0	0
ECG (6)	Average of Accuracy	**0.968**	0.960	0.963	0.961
	Average of Ranking	**1.000**	2.333	2.500	2.500
	Number of Best Accuracy	**6**	2	2	1
Image (29)	Average of Accuracy	**0.887**	0.859	0.866	0.869
	Average of Ranking	**1.379**	3.034	2.586	2.724
	Number of Best Accuracy	**22**	4	3	2
Motion (14)	Average of Accuracy	**0.806**	0.783	0.788	0.795
	Average of Ranking	**1.857**	2.857	2.429	2.071
	Number of Best Accuracy	**7**	4	3	4
Sensor (18)	Average of Accuracy	**0.878**	0.862	0.861	0.869
	Average of Ranking	2.000	2.222	2.556	**1.833**
	Number of Best Accuracy	**11**	6	4	9
Simulated (5)	Average of Accuracy	**0.991**	0.965	0.971	0.989
	Average of Ranking	2.000	1.800	2.600	**1.600**
	Number of Best Accuracy	**3**	**3**	2	**3**
Spectro (7)	Average of Accuracy	**0.941**	0.900	0.886	0.897
	Average of Ranking	**1.286**	2.000	3.143	2.000
	Number of Best Accuracy	**5**	4	1	3
Total Average (85)	Average of Accuracy	**0.879**	0.855	0.857	0.863
	Average of Ranking	**1.576**	2.612	2.682	2.341
	Number of Best Accuracy	**59**	23	14	21
	Processing Time [s]	1895.194	**995.820**	1020.266	1421.505
	Number of Parameters	273,619	**10,859**	16,215	27,067

(average accuracy of 0.855, Average of Ranking of 2.224). In terms of the number of instances with the highest accuracy, the PMC-MHLITE model achieved the highest accuracy on approximately twice as many data points (45 instances) as the LITE model (24 instances), demonstrating that the proposed method is a highly versatile model.

E. Evaluation Experiment 2

Table **??** provides a comprehensive comparison of the performance of each model across all 85 datasets. We discuss the trade-off between the effectiveness and computational cost of the proposed model based on these results.

First, in terms of accuracy, the MC-MHLF model recorded an average correct rate of 0.879 and an Average of Ranking of 1.576, demonstrating the best performance among the four models. However, the MC-MHLF model has a notably high number of parameters—approximately 274,000—and the longest average processing time at 1,895.194 seconds. In contrast, the MC-MHLITE model, which was designed to be more lightweight, reduced the number of parameters to approximately one-seventeenth (16,215) of that of the MC-MHLF model, while still achieving an average accuracy of 0.857, surpassing the baseline LITE model (0.855).

Furthermore, focusing on the PMC-MHLITE model, the average accuracy improved to 0.863, and the Average of Ranking also improved to 2.341. Although the number of parameters has increased slightly to 27,066, it remains about one-tenth the size of the MC-MHLF model. In terms of processing time, while the MC-MHLF model takes approximately 1,900 seconds, the PMC-MHLITE model takes about 1,422 seconds, achieving a reduction of approximately 25

F. Evaluation Experiment 3

Table **??** shows the results of comparing the model training and inference times for the MC-MHLF, LITE, MC-MHLITE, and PMC-MHLITE models using datasets selected for each category.

Since the processing times were generally not very long, although some errors may have occurred, the results generally aligned with the number of model parameters.

TABLE III: Comparison of Computational Costs Across Models

Dataset	Metric	MC-MHLF	LITE	MC-MHLITE	PMC-MHLITE
Adiac	Average learning time [s]	1608	**1062**	1075	1561
	Average inference time [s]	0.067	**0.060**	0.061	0.069
CBF	Average learning time [s]	753	**667**	675	873
	Average inference time [s]	0.078	**0.068**	0.072	0.084
Coffee	Average learning time [s]	**385**	398	400	511
	Average inference time [s]	0.053	**0.051**	0.052	0.057
Computers	Average learning time [s]	567	**403**	427	595
	Average inference time [s]	0.081	**0.058**	0.064	0.074
ECG200	Average learning time [s]	**341**	388	389	483
	Average inference time [s]	0.052	**0.050**	0.051	0.058
Gun_Point	Average learning time [s]	523	501	**461**	667
	Average inference time [s]	0.059	**0.054**	0.056	0.061
wafer	Average learning time [s]	8813	**5441**	5816	7830
	Average inference time [s]	0.245	**0.143**	0.162	0.234

TABLE IV: Comparison of classification accuracy for each model in the “ Device ” category

Dataset	MC-MHLF	LITE	MC-MHLITE	PMC-MHLITE
Computers	0.816	**0.824**	0.784	0.792
ElectricDevices	**0.761**	0.714	0.707	0.712
LargeKitchen Appliances	**0.912**	0.891	0.888	0.877
Refrigeration Devices	**0.589**	0.576	0.579	0.587
ScreenType	**0.621**	0.603	0.616	0.595
SmallKitchen Appliances	**0.832**	0.800	0.792	0.784
Average of Accuracy	**0.755**	0.735	0.728	0.724
Average of Ranking	**1.167**	2.333	3.167	3.333
Number of Best Accuracy	**5**	1	0	0

VI. Conclusion

In this paper, with the aim of achieving both high accuracy and lightweight performance in TSC, we propose the MC-MHLITE and PMC-MHLITE models, which integrate the MACD histogram—designed to capture dynamic changes in time series—into the lightweight LITE model, which utilizes depth-wise separable convolutions. By introducing short- and long-term MACD histograms that represent the acceleration of time series data, we enabled the efficient learning of trend changes that are difficult to extract from a single time series alone.

VII. Acknowledgments

This work was supported by JSPS KAKENHI Grant Number 23K11340. The authors wish to acknowledge Professor Eamonn Keogh and all those who contributed to the creation of the UCR time series classification archive.

Appendix

TABLE V: Comparison of classification accuracy for each model in the “ ECG ” category

Dataset	MC-MHLF	LITE	MC-MHLITE	PMC-MHLITE
ECG200	**0.930**	0.910	**0.930**	0.910
ECG5000	**0.948**	**0.948**	0.946	0.945
ECGFiveDays	**1.000**	**1.000**	**1.000**	**1.000**
NonInvasiveFatal ECG_Thorax1	**0.962**	0.950	0.957	0.960
NonInvasiveFatal ECG_Thorax2	**0.965**	0.955	0.950	0.953
TwoLeadECG	**1.000**	0.997	0.997	0.999
Average of Accuracy	**0.968**	0.960	0.963	0.961
Average of Ranking	**1.000**	2.333	2.500	2.500
Number of Best Accuracy	**6**	2	2	1

TABLE VI: Comparison of classification accuracy for each model in the “ Image ” category

Dataset	MC-MHLF	LITE	MC-MHLITE	PMC-MHLITE
50words	**0.820**	0.745	0.787	0.778
Adiac	**0.867**	0.801	0.818	0.857
ArrowHead	**0.909**	0.777	0.823	0.829
BeetleFly	**1.000**	0.950	**1.000**	0.950
BirdChicken	0.950	**1.000**	0.950	**1.000**
DiatomSize Reduction	**0.980**	0.958	0.931	0.961
DistalPhalanx OutlineAgeGroup	**0.863**	0.835	0.845	0.808
DistalPhalanx OutlineCorrect	**0.843**	0.818	0.812	0.817
DistalPhalanxTW	**0.808**	0.775	0.793	0.790
FaceAll	**0.968**	0.856	0.921	0.927
FaceFour	**0.989**	0.955	0.920	0.943
FacesUCR	0.961	0.960	**0.963**	0.961
FISH	0.994	0.989	0.994	**1.000**
HandOutlines	0.883	**0.914**	0.892	0.890
Herring	0.750	**0.766**	0.750	0.734
MedicalImages	**0.782**	0.775	0.758	0.771
MiddlePhalanx OutlineAgeGroup	**0.813**	0.788	0.783	0.775
MiddlePhalanx OutlineCorrect	**0.848**	0.810	0.838	0.822
MiddlePhalanxTW	**0.647**	0.639	0.627	0.629
OSULeaf	**1.000**	0.950	0.996	0.992
Phalanges OutlinesCorrect	**0.848**	0.843	0.845	0.846
ProximalPhalanx OutlineAgeGroup	0.883	0.878	**0.888**	0.873
ProximalPhalanx OutlineCorrect	**0.931**	0.921	0.914	0.911
ProximalPhalanxTW	0.817	**0.827**	0.825	0.815
ShapesAll	**0.943**	0.888	0.900	0.937
SwedishLeaf	**0.989**	0.960	0.971	0.968
Symbols	**0.981**	0.963	0.972	0.972
WordsSynonyms	**0.724**	0.655	0.683	0.716
yoga	**0.935**	0.907	0.912	0.924
Average of Accuracy	**0.887**	0.859	0.866	0.869
Average of Ranking	**1.379**	3.034	2.586	2.724
Number of Best Accuracy	**22**	4	3	2

TABLE VII: Comparison of classification accuracy for each model in the “ Motion ” category

Dataset	MC-MHLF	LITE	MC-MHLITE	PMC-MHLITE
Cricket_X	0.815	**0.821**	0.787	0.818
Cricket_Y	0.831	**0.862**	0.815	0.831
Cricket_Z	0.869	0.841	0.841	**0.877**
Gun_Point	**1.000**	**1.000**	**1.000**	**1.000**
Haptics	0.591	0.565	**0.604**	0.591
InlineSkate	**0.487**	0.422	0.465	0.471
ToeSegmentation1	**0.982**	**0.982**	0.969	0.965
ToeSegmentation2	**0.962**	0.923	0.938	0.915
uWaveGesture Library_X	0.819	0.812	**0.833**	0.831
uWaveGesture Library_Y	0.733	0.716	0.741	**0.743**
uWaveGesture Library_Z	**0.775**	0.756	0.772	0.764
uWaveGesture LibraryAll	0.895	0.900	0.905	**0.911**
Worms	**0.735**	0.591	0.591	0.635
WormsTwoClass	**0.796**	0.768	0.773	0.779
Average of Accuracy	**0.806**	0.783	0.788	0.795
Average of Ranking	**1.857**	2.857	2.429	2.071
Number of Best Accuracy	**7**	4	3	4

TABLE VIII: Comparison of classification accuracy for each model in the “ Sensor ” category

Dataset	MC-MHLF	LITE	MC-MHLITE	PMC-MHLITE
Car	**0.933**	**0.933**	0.883	0.900
Chlorine Concentration	0.816	**0.826**	0.823	0.813
CinC_ECG_torso	0.903	0.871	0.946	**0.959**
Earthquakes	**0.826**	0.823	0.820	0.823
FordA	0.933	0.951	**0.955**	**0.955**
FordB	0.929	0.934	0.934	**0.939**
InsectWingbeat Sound	0.584	0.605	0.625	**0.633**
ItalyPower Demand	0.968	**0.970**	0.959	**0.970**
Lighting2	**0.885**	0.820	0.836	0.836
Lighting7	**0.877**	0.822	0.795	0.849
MoteStrain	**0.895**	0.886	0.855	0.872
Phoneme	**0.363**	0.309	0.306	0.328
Plane	**1.000**	**1.000**	**1.000**	**1.000**
SonyAIBORobot Surface	**0.982**	0.845	0.885	0.870
SonyAIBORobot SurfaceII	**0.940**	0.937	0.903	0.920
StarLightCurves	0.967	0.979	0.979	**0.980**
Trace	**1.000**	**1.000**	**1.000**	**1.000**
wafer	**0.999**	**0.999**	**0.999**	**0.999**
Average of Accuracy	**0.878**	0.862	0.861	0.869
Average of Ranking	2.000	2.222	2.556	**1.833**
Number of Best Accuracy	**11**	6	4	9

TABLE IX: Comparison of classification accuracy for each model in the “ Simulated ” category

Dataset	MC-MHLF	LITE	MC-MHLITE	PMC-MHLITE
CBF	0.996	**1.000**	0.982	0.998
MALLAT	**0.981**	0.966	0.956	0.965
ShapeletSim	**0.983**	0.861	0.917	**0.983**
synthetic_control	0.993	**1.000**	**1.000**	**1.000**
Two_Patterns	**1.000**	**1.000**	**1.000**	**1.000**
Average of Accuracy	**0.991**	0.965	0.971	0.989
Average of Ranking	2.000	1.800	2.600	**1.600**
Number of Best Accuracy	**3**	**3**	2	**3**

TABLE X: Comparison of classification accuracy for each model in the “ Spectro ” category

Dataset	MC-MHLF	LITE	MC-MHLITE	PMC-MHLITE
Beef	**0.933**	0.767	0.833	0.867
Coffee	**1.000**	**1.000**	**1.000**	**1.000**
Ham	0.819	**0.838**	0.743	0.752
Meat	0.983	**1.000**	0.917	0.900
OliveOil	**0.967**	0.900	0.933	**0.967**
Strawberry	**0.980**	**0.980**	0.979	**0.980**
Wine	**0.907**	0.815	0.796	0.815
Average of Accuracy	**0.941**	0.900	0.886	0.897
Average of Ranking	**1.286**	2.000	3.143	2.000
Number of Best Accuracy	**5**	4	1	3